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SYSTEMS AND METHODS FOR FLEXURE-BASED STANDALONE INTERBODY FUSION

Abstract

Systems and methods for providing a flexure-based standalone interbody fusion device are described. In some implementations, a flexure-based standalone interbody fusion device includes one or more endplates (e.g., a first endplate and a second endplate). In some implementations, the device includes a flexure assembly (e.g., disposed between the two endplates), which in some cases is configured to bear at least a portion of the load applied to the endplates. Some iterations of the flexure assembly include one or more flexures. In some cases, applying a compressive force to the endplates places tension on the flexure, thereby providing a device that is flexible and resistant to buckling and breaking. Additional implementations are described.

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Background/Summary

RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/556,852 (Attorney Docket No. 23845.178), filed Feb. 22, 2024, and entitled SYSTEMS AND METHODS FOR FLEXURE-BASED STANDALONE INTERBODY FUSION; the contents of which are incorporated herein by reference in their entirety.

FIELD

[0002] The present disclosure relates to medical implants, and more particularly to spinal implants for interbody fusion.

BACKGROUND

[0003] One significant cause of back pain stems from spinal instability. Chronic spinal instability can lead to modic changes, including bone changes that lead to nerve compression. Nerve decompression is often treated by surgical intervention. In many cases, spinal surgery to decompress nerves (such as disc replacement surgery) requires the removal of stabilizing structures, such as the anterior longitudinal ligament, the posterior longitudinal ligament, and the annulus. Thus, attempts to relieve back pain (or correct other spinal issues) can lead to increased spinal instability and increased pain.

[0004] Interbody fusion devices have been developed to restore spinal stability by providing a rigid connection between adjacent vertebral bodies. However, if a spinal segment is made too rigid, it can result in deleterious hypermobility at adjacent spinal segments, and thus cause adjacent level degeneration. Furthermore, immediately after implantation there is often significant motion at the bone-to-implant interface, which can be painful and can inhibit bony attachment. Ultimately, stability at the operative spinal level can rely on relatively slow formation of fibrous tissue in the bone-to-implant interface; after which hypermobility can be caused at the adjacent levels.

[0005] In an attempt to counteract some of these issues, such as to prevent adjacent level iatrogenic hyper-mobility, and subsequent adjacent level degeneration, disc replacement devices have been created. Some disc replacement devices provide motion at the operative level and in turn avoid hypermobility at the adjacent levels. However, some existing disc replacement devices are not sufficiently stable and inappropriately recruit surrounding tissues to provide spinal stability. This is often exacerbated by surgically overstuffing the disc space with an implant that is too tall and thus stretching the surrounding tissues beyond their healthy limits to attempt to restrain motion. Accordingly, this can place undue strain on soft tissues, which can damage the tissues (e.g., through excessive stretching). The result can be heterotopic ossification (e.g., bone generation outside of the skeleton), with bone bridging across the disc space, outside the disc space.

[0006] Many disc replacement devices rely on sliding components. For example, some devices rely on a ball-in-socket mechanism. Some sliding mechanisms are not inherently stable and do not effectively restore spinal stability. Also, some sliding components generate wear debris (e.g., material grinding or flaking off due to the friction forces generated by sliding over time). The debris from the frictional forces can cause osteolysis (bone degradation) and concomitant problems. In some cases, such devices are encased within a sheath to prevent release of wear debris, but some such sheaths are themselves susceptible to degradation, which can result in bulk release of wear debris upon rupture of the sheath.

[0007] Some disc replacement devices rely on polymeric cores to provide stable motion. Polymers can be flexible. However, existing polymeric devices do not appropriately restore stability in all planes of motion. For example, the polymer might provide appropriate flexion/extension stiffness but consequently does not provide appropriate stiffness in lateral bending, axial rotation, nor axial compression. Also, polymers oxidize within the body which changes the stiffness of the polymer. Furthermore, polymers have finite fatigue life and will ultimately fail from wear.

[0008] Some devices have decreased rigidity in order to generate additional flexibility, but this can lead to buckling, bending, fatigue, and other undesirable effects on the device. In some devices—in an attempt to make interbody devices that are flexible—yet rigid enough to prevent buckling—carcinogenic materials have been incorporated, which can lead to even more severe medical problems (e.g., cancer) down the road. Moreover, due to their design, many devices are incapable of achieving a desired level of flexibility while still having a desired level of stability, regardless of the type of material used.

[0009] Thus, while techniques currently exist that are used to provide stabilizing spinal devices, challenges still exist, including those listed above. Accordingly, it would be an improvement in the art to augment or even replace current techniques with other techniques.

SUMMARY

[0010] It is often desirable for an interbody device (i.e., an interbody fusion device for standalone interbody fusion) to have a load-bearing component that is sufficiently rigid so as to stabilize the spine and to provide the structural support needed to allow the spine to heal and to minimize pain. However, it is also generally desirable for the device to provide enough flexibility on one or more axes, including a flexion/extension axis, a lateral bending axis, and a torsion axis (also called an axial rotation axis) to prevent overloading adjacent spinal levels. If the device is not sufficiently flexible, the spinal levels adjacent to the treated level can often be forced into hypermobility, which can lead to accelerated degeneration, or otherwise cause pain and damage to the patient (which can be long-term or even permanent). Some implementations of the present systems and methods provide an interbody device that is both stable and flexible.

[0011] According to some implementations, the interbody device includes one or more endplates (e.g., a first endplate and a second endplate) configured to be disposed adjacent to the natural bony endplates of spinal vertebrae. Additionally, some implementations include one or more flexure assemblies disposed between the endplates and configured to provide stability to the implant when a load is placed on the endplates (e.g., the compressive force applied by the vertebrae between which the implant is disposed). Moreover, some implementations of the flexure assembly are also configured to be flexible (e.g., to mimic one or more physiological conditions of a healthy spine).

[0012] In some iterations of the systems and methods disclosed herein, the desired stability and flexibility of the flexure assembly is achieved through the provision of one or more tension-based components, or flexures. Indeed, by providing a tension-based load-bearing component as opposed to a compression-based one, a load can be supported and flexibility can be maintained with a greatly decreased risk of components buckling, fatiguing, or breaking. Accordingly, the described flexure can include any suitable component capable of being placed under tension, and the flexure assembly can include any suitable components (in any suitable configuration) for taking the compressive force applied to the implant endplates and transposing it to a tensile force in the flexure.

[0013] In some implementations, the flexure assembly includes one or more loading regions (e.g., for receiving the compressive force) and one or more supports (e.g., for transferring the load from the loading regions to the flexure). For example, a first support can transfer a first portion of a compressive force from a first loading region to a first end of the flexure, and a second support can transfer a second portion of the compressive force from a second loading region to a second end of the flexure. In some cases, the first end of the flexure is positioned opposite to the first loading region, and the second end of the flexure is positioned opposite to the second loading region (e.g., the first end of the flexure is farther from the first loading region than is the second end of the flexure, and the second end of the flexure is farther from the second loading region than is the second end of the flexure). This can allow the flexure (or any suitable portion thereof) to be placed under tension rather than under compression, thus allowing it to have a sligher aspect ratio and better flexibility and stability properties than some compression-based components.

[0014] Although in some cases a single flexure assembly is sufficient, some implementations

include multiple flexure assemblies. In some implementations, multiple flexure assemblies with specific flexibility profiles have certain orientations allowing them to sum and work together to provide the desired characteristics of the implant as a whole (e.g., specific flexibilities along the flexion/extension axis, the lateral bending axis, and the torsion axis). As many different combinations are possible, additional details are discussed below.

Description

DESCRIPTION OF THE FIGURES

[0015] The objects and features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying figures. Understanding that these figures depict only typical embodiments of the disclosed systems and methods and are, therefore, not to be considered limiting of its scope, the systems and methods will be described and explained with additional specificity and detail through the use of the accompanying figures in which:

[0016] FIG. 1 shows a perspective view of an interbody implant, as known in the art;

[0017] FIG. 2 shows a perspective view of a flexure-based interbody implant, in accordance with a representative embodiment of the disclosed systems and methods;

[0018] FIG. 3 shows a perspective view of a stem assembly, in accordance with some embodiments;

[0019] FIG. 4 shows a perspective view of a flexure assembly, in accordance with a representative embodiment;

[0020] FIG. 5 shows a perspective view of the flexure assembly, in accordance with a representative embodiment;

[0021] FIGS. 6A-6D show perspective views of a plurality of flexure assemblies coupled together in differing configurations, in accordance with some representative embodiments;

[0022] FIG. 7 shows a perspective view of the flexure assembly, in accordance with a representative embodiment;

[0023] FIG. 8 shows a plan view of the flexure assembly, in accordance with a representative embodiment;

[0024] FIG. 9 shows a top perspective view of the flexure-based interbody implant, in accordance with a representative embodiment;

[0025] FIG. 10 shows a plan view of the flexure-based interbody implant, in accordance with a representative embodiment;

[0026] FIG. 11 shows a front elevation view of the flexure-based interbody implant, in accordance with a representative embodiment;

[0027] FIG. 12 shows a side elevation view of the flexure-based interbody implant, in accordance with a representative embodiment;

[0028] FIG. 13 shows a perspective view of the flexure assembly arrangement for the flexure-based interbody implant, in accordance with a representative embodiment;

[0029] FIG. 14 shows a perspective view of the flexure-based interbody implant with a porous structure, in accordance with a representative embodiment;

[0030] FIG. 15 shows a cutaway view of the flexure-based interbody implant with the porous structure, in accordance with a representative embodiment;

[0031] FIG. 16 shows a perspective view of the flexure-based interbody implant with locking caps disposed in secondary fixation passages, in accordance with a representative embodiment; and

[0032] FIG. 17 shows a front elevation view of the flexure-based interbody implant installed in an interbody space between two spinal vertebrae, in accordance with a representative embodiment.

DETAILED DESCRIPTION

[0033] A description of embodiments will now be given with reference to the figures. It is expected that the present systems and methods may take many other forms and shapes. Hence, the following disclosure is intended to be illustrative and not limiting, and the scope of the disclosure should be determined by reference to the appended claims.

[0034] Systems and methods for providing a standalone interbody device are disclosed. Generally speaking, it is often desirable for an interbody device to have a load-bearing component that is sufficiently rigid so as to stabilize the spine and to provide the structural support needed to allow the spine to heal and to minimize pain. However, it is also generally desirable for the device to provide flexibility on or more axes, including a flexion/extension axis, a lateral bending axis, a torsion axis (also called an axial rotation axis), and any combination thereof. Indeed, in some embodiments, it is desirable for the device to provide flexibility in each of the flexion/extension axis, the lateral bending axis, and the torsion axis. If the device is not sufficiently flexible, this can (in some cases) overload adjacent spinal levels, limit a patient's mobility, cause friction between spinal vertebrae and the device, and cause pain and damage to the patient (which can be long-term or even permanent). Thus, some embodiments of the present systems and methods provide an interbody device that is both stable and flexible.

[0035] Some implant makers have attempted to provide a stable and flexible device by including one or more sliding components, but such components are not always stable and often generate a substantial amount of wear debris (e.g., from material flaking or rubbing off due to the friction forces generated by sliding over time). The debris and the frictional forces can cause osteolysis (or bone degradation) and other major problems. In some cases, such interbody devices are encased within a sheath to prevent release of wear debris, but some such sheaths are themselves susceptible to degradation, which can result in bulk release of wear debris upon rupture of the sheath.

Moreover, sliding components can wear out over time, thereby requiring eventual replacement. Some embodiments of the present systems and methods provide an interbody device that does not have sliding components, thereby averting the aforementioned problems.

[0036] Some interbody devices attempt to provide flexibility without sliding components by utilizing a compression-based flexible member sandwiched between contact structures. For example, FIG. 1 shows an interbody device **50** having a first contact structure **52a** and a second contact structure **52b**, with a stem **54** spanning the gap between the two contact structures, as taught in the prior art. This configuration can have several substantial drawbacks. For one, the stability of the device can increase proportionally to the thickness of the stem, but the flexibility of the device can decrease proportionally to the thickness of the stem. Accordingly, stability and flexibility can have an inversely proportional relationship in such devices (and cannot, generally, both be increased together). Thus, when the stem is thin enough to be sufficiently flexible, it can also be weaker, and therefore it can be highly susceptible to fatigue failure, buckling and breaking. Furthermore, some compression-based flexible structures do not provide anatomical stiffness simultaneously in all planes of motion. For example, a human cervical disc is relatively incompressible but quite flexible in flexion/extension, whereas a helical coil spring is often a compression-based flexible structure that has a compression stiffness directly related to its flexion/extension stiffness.

[0037] Some embodiments of the present systems and methods address one or more of the foregoing issues by providing a flexure-based standalone interbody implant **100**. The implant can include any component suitable for use with an interbody device, and in particular an interbody device that has one or more flexure assemblies **110** for providing stability and flexibility to the implant. For example, FIG. 2 shows an embodiment of a flexure-based interbody implant **100** having a first endplate **102a** and a second endplate **102b** (each optionally having a flange **104a/104b** with a primary fixation passage **106a/106b** and a secondary fixation passage **108a/108b** formed therethrough), with the first endplate **102a** being separated from the second endplate **102b** by one or more flexure assemblies (e.g., a first flexure assembly **110a** and a second flexure

assembly **110b**), as opposed to using a single stem (such as the stem **54** of FIG. **1**).

[0038] Using one or more flexure assemblies **110** (with one or more tension-based flexures **114** as opposed to a single compression-based stem **54** operating as a load-bearing component) can provide one or more features, so various flexure assembly configurations—as well as other implant elements utilized in some embodiments—are discussed below. In some cases, this includes providing flexure geometry that can restore appropriate stability in each and every plane of spinal motion (e.g. flexion/extension, lateral bending, axial rotation, and compression). The ranges of stiffness for each plane of spinal motion can be unique to that plane, and can vary for each level of the spine. Independently and simultaneously matching each stiffness, is a feature of some embodiments of the described flexure geometry that is not found in most, if any, competing devices. Moreover, a flexible interbody device that specifically and simultaneously matches stiffness for each plane (or specific planes) of spinal motion and then also has a graft window (as provided by some embodiments described herein) is also a feature that is not found in most, if any, competing devices.

[0039] The described implant **100** can include any suitable component or characteristic that allows it to function as described herein. Indeed, in some embodiments, the implant includes one or more endplates **102** (such as a first endplate **102a** and a second endplate **102b**). The endplate can include any suitable component configured to be placed adjacent to, align with, couple with, or otherwise interact with the natural bony endplate of a patient's spinal segment. For example, the endplates can include one or more plates, sheets, frames, surfaces, planes, load-bearing regions, scaffolds, surfaces, contacts, porous surfaces, or other components configured to act as an artificial endplate or to interface with a natural bony endplate. By way of non-limiting illustration, FIGS. **2**, **9**, **12**, **14**, and **16** show some embodiments of implants **100** that each have a first endplate **102a** and a second endplate **102b**, wherein each of the endplates include a substantially planar plate or frame configured to be disposed adjacent to a natural bony endplate of a spinal segment. Generally, bio-compatible materials that have been made rough provide suitable materials for boney ingrowth.

[0040] The endplate **102** can have any shape suitable for interfacing with a natural endplate. For example, some embodiments of the endplate are (or are substantially) circular, semi-circular, triangular, square, rectangular, trapezoidal, pentagonal, hexagonal, star-shaped, T-shaped, polygonal, rounded, symmetrical, asymmetrical, shaped to match or otherwise correspond with a natural endplate, or any other regular or irregular shape capable of interfacing with a natural endplate. By way of non-limiting illustration, FIGS. **9-10** show some embodiments in which the endplates **102a** and **102b** are substantially rectangular or quadrilateral in shape. Generally, maximizing contact area can minimize contact stress to provide for bone health. Also, filling the intervertebral space to the perimeter of each endplate can help provide for contacting the strongest portion of the vertebral endplate.

[0041] Some embodiments of the endplate **102** are solid or unbroken, while others are broken or discontinuous. Moreover, some embodiments are smooth, while others are knurled, porous, perforated, roughened, undulated, recessed, provided with one or more protrusions, or are otherwise textured. Indeed, some embodiments of one or more endplates are flat, while others are contoured. By way of non-limiting illustration, FIGS. **2**, **9**, **12**, **14**, and **16** show some embodiments of implants **100** that each have a first endplate **102a** and a second endplate **102b** that each include a substantially planar plate or frame with one or more gaps (e.g., windows, as discussed in more detail below) formed therein.

[0042] Where the described implant **100** includes multiple endplates **102**, the endplates can run parallel or substantially parallel with respect to each other when the implant is at rest, or they can be angled with respect to each other when the implant is at rest. By way of example, FIG. **12** shows an embodiment of the implant **100** with its first endplate **102a** and its second endplate **102b** being slightly angled such that they are slightly farther apart at a ventral side (near flanges **104a** and **104b**) than at a dorsal side.

[0043] According to some embodiments, the implant **100** includes one or more primary fixation mechanisms. The primary fixation mechanism can include any component configured to secure the implant in the proper location within the patient. For example, the primary fixation mechanism can include any component configured to secure the implant in an intervertebral space (e.g., by securing it to one or more spinal vertebrae).

[0044] In some embodiments, the primary fixation mechanism includes one or more anchors, catches, flanges, fasteners, burrs, barbs, frictional engagements, mechanical engagements, and any other suitable feature that can be used to secure the implant **100** in an intervertebral space. Indeed, in some embodiments, the primary fixation mechanism comprises one or more flanges **104**.

[0045] Where the primary fixation mechanism includes a flange, the flange can include any suitable component that is configured to receive one or more anchors **107** or to couple to a portion of a patient (e.g., to a spinal vertebra). The flange can also be any suitable shape, including being substantially semi-circular, square, rectangular, triangular, trapezoidal, pentagonal, hexagonal, otherwise polygonal, symmetrical, asymmetrical, or any other regular or irregular shape suitable for receiving an anchor. By way of non-limiting illustration, FIG. **11** shows an embodiment, in which the flange **104a** is somewhat quadrilateral and asymmetrical in shape.

[0046] The flange **104** can also be coupled to the endplate **102** or any other portion of the implant **100** in any suitable manner, such as via one or more nails, screws, bolts, staples, eyelets, adhesives, welds, interference fits, friction fits, tongue-and-groove connections, snaps, ties, rivets, or any other coupling mechanism, or by being integrally formed with one or more components of the implant. Moreover, the flange can be coupled at any angle, including parallel, perpendicular, acute, obtuse, at an intersecting angle with respect to a corresponding endplate, or any other suitable configuration or combination thereof. By way of non-limiting illustration, FIG. **11** shows an embodiment of the implant **100** with a first flange **104a** and a second flange **104b**, with the first flange being integrally formed with a first endplate **102a**, and the second flange being integrally formed with a second endplate **102b**, with each flange having a somewhat of a rounded, polygonal shape with a substantially rectangular or quadrilateral portion in which one or more primary fixation passages **106a** and **106b** are formed, and a substantially triangular portion or sloped portion in which one or more secondary fixation passages **108a** and **108b** are formed. In FIG. **11**, each flange is substantially perpendicular to its respective endplate, and is coupled to each respective endplate at a ventral edge of the endplate.

[0047] As briefly mentioned in the example in the preceding paragraph, some embodiments of the implant **100** include one or more primary fixation passages **106**. The primary fixation passage can include any suitable feature that it is configured to couple with one or more anchors **107**. Moreover, some embodiments include one or more secondary fixation passages **108**. The secondary fixation passage can include any feature that it is configured to couple to one or more secondary anchors or any other suitable component for further securing the anchor **107** in place, such as a locking cap **109** that is configured to selectively lock a primary anchor in place. By way of non-limiting illustration, FIG. **11** shows an embodiment of the implant **100** with flanges **104a** and **104b**, each of which has a primary fixation passage **106a** and **106b** and a secondary fixation passage **108a** and **108b**. In FIG. **11**, the primary fixation passages **106** are substantially circular, and are formed in a somewhat rectangular portion of each of the respective flanges **104**. The secondary fixation passages **108** in FIG. **11** are also substantially circular, are optionally smaller than the primary fixation passages **106**, and are formed in a sloped, angled, or tapered portion of the respective flanges **104**. Similar passages **106** and **108** can also be seen in the implant **100** of FIG. **14**.

[0048] As mentioned, some embodiments of the implant **100** include one or more anchors **107**. Where the implant includes an anchor, the anchor can include any suitable anchoring component for securing the implant in place, such as one or more screws, bolts, threaded couplings, nails, stakes, pins, staples, pawls, spikes, blades, barbs, catches, or any other suitable anchors. By way of non-limiting illustration, FIG. **17** shows an embodiment of the implant **100** with a first anchor **107a**

in the form of a bolt or screw placed through a first primary fixation passage **106a**, and a second anchor **107b** in the form of a bolt or screw placed through a second primary fixation passage **106b**, thereby securing the implant to a patient's spinal vertebrae.

[0049] Some embodiments of the implant **100** include one or more secondary securement components. Where the implant includes a secondary securement component, the secondary securement component can include anything that helps the anchors **107** remain fixed in place, such as one or more secondary anchors, locks, caps, or other securement components. For example, some embodiments include one or more locking caps **109** configured to selectively cover at least a portion of a head of the anchor to prevent the anchor from backing out. Where the implant includes a locking cap, the locking cap can have any component or configuration that allows it to help prevent an anchor from backing out. For example, in some embodiments the locking cap has a lip or protection that is configured to selectively cover all or a portion of the anchor (e.g., by covering all or a portion of the primary fixation passage **106**). In some embodiments, the locking cap is rotatable with respect to the secondary fixation passage **108** such that the lip can be rotatably disposed in front of the primary fixation passage (thereby blocking an anchor from being inserted or removed therethrough), or rotatably disposed in an unlocked position wherein the lip does not cover the primary fixation passage, thereby allowing an anchor to be inserted or removed. In some embodiments, the locking cap is configured to lock in place in the deployed position (e.g., by snapping, hooking, engaging with a locking component, or otherwise locking in place). By way of non-limiting illustration, FIG. **16** shows an embodiment of the implant **100** with a first locking cap **109a** placed through a first secondary fixation passage **108a** and a second locking cap **109b** placed through a second secondary fixation passage **108b**, wherein the locking caps are configured to be rotated to cover a portion of the respective primary fixation passages (and thereby prevent anchors from backing out of said primary fixation passages).

[0050] Regarding general characteristics of the implant **100**, some embodiments of the implant provide a desired range of flexibility along one or more of a flexion/extension axis, a lateral bending axis, and an axial rotation or torsion axis. Though it does not necessarily occur in all embodiments, in some embodiments, the amount of flexibility in each of these planes matches that of the adjacent healthy spinal levels so as to restore stability without making the segment too rigid. In some embodiments, the desired range of flexibility closely matches the physiological range (although the flexibility of the implant can also be greater or lesser than physiological flexibility, if desired). In some embodiments, the desired range of flexibility depends on the target level of the spine (or other area of the body in which the implant will be used), and it may vary between spinal segments of a patient or across patients. In accordance with some embodiments, the desired degree of flexibility can be anywhere between 0 degrees and 20 degrees, or any subrange thereof (e.g., between 1 and 15 degrees, between 2 and 10 degrees, or any other desirable subrange). By way of non-limiting illustration, in some embodiments the flexion/extension flexibility of the implant is 8 degrees $\pm$ 3 degrees for cervical spine implementations, or 6 degrees $\pm$ 3 degrees for lumbar spine implementations (or 7 degrees $\pm$ 4 degrees generally). In some embodiments, the lateral bending flexibility of the implant is 6 degrees $\pm$ 2 degrees for cervical spine implementations, or 4 degrees $\pm$ 2 degrees for lumbar spine implementations (or 5 degrees $\pm$ 3 degrees generally). In some embodiments, the axial rotation flexibility is 5 degrees $\pm$ 3 degrees for cervical spine implementations, or 2 degrees $\pm$ 2 degrees for lumbar spine implementations (or 4 degrees $\pm$ 3.5 degrees generally). In some embodiments, the range of motion for a particular spinal segment of a particular patient is measured prior to operation (e.g., using preoperative X-rays, MRI, CT scan, or any other suitable imaging technique) and the flexibility of the implant is tailored to match the measured levels of motion (or to simulate ideal levels of motion). For example, any of the foregoing values can be adjusted by an amount between 0.1 degrees and 2 degrees, or any subrange thereof (e.g., between any two of the following: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.25, 1.5, 1.75, 2, or any other value that is less than two degrees in order to achieve the desired levels of

flexibility).

[0051] In some embodiments, the implant **100** has a certain compression stiffness while having a different, lower stiffness in bending. The compression stiffness can be any stiffness suitable for a medical implant, such as any stiffness desirable for insertion between two adjacent vertebrae of a patient's spine and is, in some embodiments, made higher than the minimum compression stiffness of an intervertebral disc. Though the implant can have any suitable bending stiffness, in some embodiments, the bending stiffness is less stiff than the compression stiffness and is stiff enough to restore equal motion to the adjacent, healthy levels. Specifically, in some embodiments, the compression stiffness of the device is at or above the compressive stiffness of an intervertebral disc while the bending stiffness is at (or similar to) the bending stiffness of a healthy intervertebral disc, which can be measured in flexion/extension with X-rays of the adjacent spinal levels. In some cases, being relatively stiff in compression while being less stiff in flexion/extension is a feature that differentiates the described systems and methods from most, if not, all other competing devices. Furthermore, it is believed that having a different stiffness in each plane of bending is a feature of some embodiments of the described systems and methods that is not found in most, if any, competing devices, noting that such feature can allow the operative level to be restored to healthy levels in each plane of motion. In this regard, some competing devices have been too flexible in flexion/extension or too flexible in axial compression or too flexible in axial rotation, etc. On this point, getting all of the stiffnesses right, at the same time is a feature of some embodiments of the described implant. In some embodiments, the compression stiffness of the implant (or a portion thereof) is between 100 MPa and 120 GPa (or any subrange thereof)—allowing the implant to protect exiting nerve roots from being compressed and noting that approximately 120 GPa is the stiffness of solid titanium, which is a desired material for some embodiments of this implant.

[0052] In some embodiments, the bending stiffness of the implant **100** is closer to the range of between 20 deg/N-M and 0.25 deg/N-M (or any subrange thereof, such as 4 ± 2 deg/N-M or any other suitable subrange). For example, some embodiments of the implant have a compressive stiffness of between: 100 MPa and 2 GPa, 200 MPa and 1.5 GPa, 300 MPa and 1.2 GPa, 500 MPa and 850 MPa, or any other subrange between 100 MPa and 120 GPa, while simultaneously having a bending stiffness of between: 0.25 deg/N-M and 5 deg/N-M, 3 deg/N-M and 5 deg/N-M, 1 deg/N-M and 8 deg/N-M, or any other subrange between 0.25 deg/N-M and 20 deg/N-M. According to some embodiments, the stiffness is sufficient for the implant (or one or more load-bearing components of the implant, as discussed below) to withstand a certain compressive force (e.g., the compressive force applied by the adjacent vertebrae of a patient's spine when the implant is inserted between such vertebrae) without compression of exiting nerve roots and while still allowing stable flexion/extension, lateral bending, and axial rotation at (or near) physiologic loads.

[0053] To help ensure that desired flexibility and stability parameters can be achieved, some embodiments of the implant **100** include one or more flexure assemblies **110**. Generally speaking, some compression-based flexible elements are unstable, whereas some tension based flexible elements can be substantially more stable. To illustrate, an object suspended from a rope hanging from the ceiling is (in some cases) much less likely to fall than an object balanced on a pole standing on its end on the floor. Moreover, in many cases, it is a lot easier to bend or break a rod by compressing it from its ends than to bend or break the rod by pulling on its ends directly outward. By way of example, FIG. 3 shows a stem assembly **60** having one or more load-bearing regions **62a/62b** separated by one or more stems **64**. When pressure is exerted on one or more of the load-bearing regions **62**, the stem is liable to fail in fatigue, buckle, collapse, or break (especially when undergoing flexing along one or more axes). In contrast, FIG. 4 shows an embodiment of a flexure assembly **110** having one or more load bearing regions **112a/112b** separated by one or more tension-based flexures **114**. In this embodiment, the flexure **114** is supported by a first support **116a** and a second support **116b**. When pressure is exerted on one or more of the load-bearing regions, the flexure **114** is placed under tension and is not likely to bend and break, even if it undergoes

significant flexing along any or all of the flexion/extension axis, lateral bending axis, or torsion axis. Moreover, as shown in FIG. 5, the stability of the flexure assembly **110** can, in some embodiments, be increased without sacrificing flexibility by increasing the thickness (or rigidity) of the supports **116** but keeping the flexure **114** relatively small or flexible.

[0054] The flexure assembly **110** can be configured to provide any desired degree of flexibility to the implant **100**. As discussed above, some embodiments of the implant are configured to provide a certain degree of flexibility along a certain axis dependent on the desired target location of the implant within the spine. Accordingly, the flexure assembly can have any suitable flexibility that allows it to impart the desired flexibility to the implant (either alone, or in combination with other flexure assemblies). For example, some embodiments of the flexure assembly are configured to provide between 0 degrees and 20 degrees of flexing (or any subrange thereof) along any particular axis. For example, some embodiments are configured to provide 2 ± 2 , 3 ± 2 , 4 ± 3 , 5 ± 3 , 6 ± 2 , 6 ± 3 , 8 ± 3 , 7 ± 5 , 10 ± 4 , at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, or at least 9 degrees, or any other suitable degree of flexing along each or any of the flexion/extension axis, the lateral bending axis, or the axial rotation axis.

[0055] To illustrate how the flexibility of the flexure assembly **110** can contribute to the flexibility of the implant as a whole, if a flexibility of 8 degrees $\pm$ 3 degrees along the flexion/extension axis is desired, and if the implant has only one flexure assembly, then such flexure assembly can have a flexibility of 8 degrees $\pm$ 3 degrees along the flexion/extension axis. If the implant has two or more flexure assemblies, then the flexure assemblies can be more flexible along the flexion/extension axis of the implant such that the cumulative flexibility of the implant along the flexion/extension axis is 8 degrees $\pm$ 3 degrees.

[0056] Similarly, the flexure assembly **110** can have any suitable stiffness as needed to contribute to the overall stiffness of the implant **100** (as discussed above). For example, where a single flexure assembly is used, the flexure assembly can have a stiffness of up to 100% of the stiffness of the implant (or of a portion of the implant that is disposed between two endplates **102**), whereas some embodiments of a flexure assembly contribute to only a portion of the stiffness of the implant (e.g., where additional flexure assemblies, compression-based elements, or other components contribute to the stiffness of the implant). That said, in some embodiments, the flexure assembly (or any of its constituent components, such as the flexure **114**, as discussed in more detail below) is configured to withstand at least 10% of the compressive force to which the implant will be subject (e.g., at least 25%, at least 50%, at least 75%, or even up to 100% of the compressive force). Thus, although the implant can have any number of flexure assemblies, a single flexure assembly (or a relatively small number, such as between 1 and 5, or fewer than 10) is sufficient, in some embodiments, to withstand the forces to which the implant will be subjected.

[0057] In line with the foregoing, the flexure assembly **110** can have any suitable configuration allowing it to impart the desired flexibility and stability to the implant **100**. In this regard, and as illustrated by FIG. 5, a flexure assembly **110** can generally be referred to as having a height **h**, a width **w**, and a length **1**. It is important to note that these designations are for reference only, and do not limit the flexure assembly to any particular orientation with respect to other portions of an implant **100**. For example, although some embodiments of a flexure assembly are configured such that the height **h** of the flexure assembly bridges the gap between the implant's endplates **102** (e.g., generally extending in a rostral-caudal direction), some embodiments are arranged in alternative configurations (e.g., such that the width **w** extends in a rostral-caudal direction, the length **1** extends in a rostral-caudal direction, a portion of the flexure assembly along a diagonal extends in a rostral-caudal direction, or the flexure assembly is disposed in an a different orientation with respect to one or more of the endplates). Notwithstanding the foregoing, for purposes of simplicity, the height **h** of a flexure assembly is generally referred to as the direction aligned substantially parallel to a central axis of the flexure, with the length **1** and the width **w** running substantially perpendicular to the central axis of the flexure. Additionally, although the width is generally

described as the direction bordered by a first support **116a** and a second support **116b** (as discussed below), the length and the width are reversed in some embodiments. Accordingly, for the purposes of this disclosure, a description of a width with characteristic A and a length with characteristic B also includes a description of a width with characteristic B and a length with characteristic A.

[0058] The flexure assembly **110** can have any suitable dimensions allowing it to perform its intended functions. In some embodiments, the flexure assembly provides a minimum height of 5 mm for cervical uses and 7 mm for lumbar uses. Moreover, it should be noted that, in some cases, mechanisms that provide stable motion but that cannot fit within a corresponding height envelope can pose the risk of overstuffing the space and damaging surrounding soft tissues and facet joints. Thus, some embodiments of the flexure assembly have a height h that equals a distance between the first endplate **102a** and the second endplate **102b** ($\pm 5\%$). That said, in some embodiments, the height of the flexure assembly is less. For example, some embodiments of the implant include two or more flexure assemblies (or other components) stacked on top of one another, such that the heights of each flexure assembly (or other components) summed together would equal the distance between the endplates ($\pm 5\%$). In some embodiments, the height of the flexure assembly is between 1 mm and 100 mm (or any subrange thereof). For example, in some embodiments, the height is between: 4 mm and 10 mm, 5 mm and 12 mm, 7 mm and 17 mm, or any other subrange between 1 and 40 mm.

[0059] Similarly, the flexure assembly **110** can have any suitable width that allows it to function as intended. In this regard, some embodiments of the flexure assembly fit within the circumference of a vertebral body. Additionally, it should be noted that cervical bodies are generally smaller than lumbar bodies. In any case, fitting within the size constraints of the index level is, in some cases, preferred. In some embodiments, the width is less than a width of the endplates (such as less than 100% of the endplate width, or less than 90%, 80%, 70%, 75%, 60%, 50%, 40%, 25%, 10%, or any other percentage of the endplate width). In some embodiments, the width of the flexure assembly is between 1 mm and 100 mm (or any subrange thereof, such as the subranges shown above with respect to height).

[0060] Similarly, some embodiments of the flexure assembly **110** have a length that is less than a length of the endplates (such as less than 100% of the endplate length, or less than 90%, 75%, 50%, 40%, 25%, 10%, or any other percentage of the endplate length). In some embodiments, the length of the flexure assembly is between 1 mm and 100 mm (or any subrange thereof, such as the subranges shown above with respect to height).

[0061] In addition to the flexure assembly **110** having any suitable dimensions, the various components of the flexure assembly, such as the loading regions **112**, the supports **116**, and the flexure **114** can have any suitable dimensions, as discussed in more detail in the descriptions of the individual components below.

[0062] The flexure assembly **110** can have any suitable shape that allows it to function as described herein, including by having one or more tension-based flexures **114**. Indeed, in some embodiments, the flexure assembly (or a portion thereof): has an N-shape, resembles a portion of the Greek fret or Greek key, resembles the shape-of an upside-down U (or first squared bracket) coupled together with a right-side-up U (or second squared bracket), is wave-shaped, comprises an undulated portion, is sinusoidal, resembles a square wave form, resembles a rectangular wave form, resembles a sawtooth waveform, resembles a triangle waveform, or has any other suitable shape that allows the flexure assembly to function as described herein. By way of non-limiting illustration, FIG. 4 shows an embodiment in which the flexure assembly **110** resembles a section of the Greek key; FIG. 11 shows an embodiment in which the flexure assembly **110** resembles a right-side-up U coupled to an upside down U with the tension-based flexure **114** comprising part of and coupling together the two U-shaped portions; and FIG. 13 shows an embodiment in which the flexure assembly **110** appears to comprise a first squared bracket in a first orientation, coupled to a second squared bracket in a second, opposite (or substantially opposite) orientation. Thus, in some

cases, the flexure assembly comprises one or more first supports **116a** (e.g., vertical, diagonal, or any other suitable supports) that are coupled to one or more tension-based flexures **144** (e.g., via one or more arched portions, transverse portions, diagonal portions, or any other suitable couplers), which in turn are coupled to one or more second supports **116b** (e.g., via one or more arched portions, transverse portions, diagonal portions, or any other suitable couplers).

[0063] The flexure assembly **110** can include any component that allows it to support a load (e.g., a load placed on the endplates **102** of the implant **100**) using tension. For example, (as mentioned above) some embodiments of a flexure assembly include one or more loading regions **112**, one or more flexures **114**, and one or more supports **116**.

[0064] For embodiments of a flexure assembly **110** that include one or more loading regions **112**, the loading regions can include any suitable component configured to receive a load, to couple with an endplate **102**, or to comprise an endplate. In some cases, as it can be desirable to avoid contact stress, a flexure assembly that avoids contact loading between components is provided. For example, some embodiments of the loading region include one or more plates, flanges, bars, bases, scaffolds, porous structures, contact surfaces, couplers, or even just a surface (of the flexure assembly or of any of its constituent components) configured to transfer a load from an endplate **102** (or from a bone) to the flexure assembly. The loading region can be any shape, including circular, semi-circular, triangular, square, rectangular, trapezoidal, pentagonal, hexagonal, star-shaped, T-shaped, L-shaped, polygonal, or any other regular or irregular shape. By way of non-limiting illustration, FIG. 4 shows an embodiment of the flexure assembly **110** with a first loading region **112a** and a second loading region **112b**, each comprising a rectangular bar or elongated member configured to receive a load. By way of further illustration, FIG. 5 shows an embodiment of the flexure assembly **110** with a first loading region **112a** and a second loading region **112b**, each comprising a square, rectangular, or polygonal plate configured to receive a load. To further illustrate, FIG. 7 shows an embodiment of the first flexure assembly **110a** with: (a) a first loading region **112a** that simply comprises an upper surface of a first support **116a** (having a generally rectangular or flat surface); and (b) a second loading region **112b** that includes a base having a larger surface area than a lower surface of a second support **116b**. FIG. 7 also shows an embodiment of the second flexure assembly **110b** with: (a) a first loading region **112a** that includes an L-shaped component coupled to the first column or support **116a** of the first flexure assembly **110a** and to the first column or support **116a** of the second flexure assembly **110b**; and (b) a second loading region **112b**, which includes a bottom surface of a second support **116b**.

[0065] It is important to note that some embodiments of the flexure assembly **110** optionally do not have a separate plate, flange, bar, or other component to act as a loading region **112**. For example, in some embodiments, the supports **116** of the flexure assemblies couple directly to the endplates **102** of the implant **100** (such that the endplates themselves act as the loading regions). In some embodiments, the supports take the load directly, such that a separate portion designated as a loading region is not necessary (instead, the ends or end portions of the supports taking the load act as the loading regions).

[0066] In some embodiments, one or more loading regions **122** of the flexure assembly **110** are configured to perform additional functions, such as to: couple other components together, define one or more passages in the implant **100** to lend additional structural support (e.g., to the endplates **102**), or otherwise improve the functionality of the implant. In accordance with some embodiments, a passage from the vertebral endplate into the disc space allows for nutrient flow to the healing mass of new tissue. On this point, it is noted that the vertebrae are generally the source of blood supply to a disc space. Thus, in some embodiments, a flexure assembly that allows for the nutrients to flow freely from the vertebral body into the disc space is provided (e.g. a graft window). By way of non-limiting illustration, FIG. 9 shows an embodiment of the first loading region **112a** of a second flexure assembly **110b** that performs some of the aforementioned functions, as it has an L-shape that allows it to define a window **126** in the implant **100**, and it connects to a first endplate

102a (thereby adding structural support).

[0067] The loading region **112** can have any suitable dimensions allowing it to bear a load. For example, some embodiments of the loading region have a length or width approximately equal to the length or width of the flexure assembly (e.g., $90\% \pm 10\%$ of a length or width of the assembly), but in some embodiments the length or width is substantially less (e.g., as low as 1% of the length or width of the flexure assembly). In many embodiments, the loading region has a height that is substantially less than the height of the flexure assembly (e.g., as shown by the loading regions **112a** and **112b** of FIG. 5). For instance, in some embodiments, the height of the loading regions is between 1% and 20% of the height of the flexure assembly, or any subrange thereof (e.g., $10\% \pm 8\%$). In some embodiments, the loading region is just a surface of a portion of the flexure assembly, and is therefore 2-dimensional and lacks a height altogether.

[0068] In some embodiments with more than one loading region **112**, the loading regions are aligned (e.g., such that the first loading region **112a** would partially or completely occlude or overlap with the second loading region **112b** when viewed from directly above), but in some embodiments, the loading regions are offset. For example, the loading regions **112a** and **112b** of the flexure assembly **110** in FIG. 4 are substantially aligned, but the loading regions **112a** and **112b** of the implant **100** in FIG. 7 are offset to such a degree that they do not overlap.

[0069] Turning now to the flexure **114** of the flexure assembly **110**, the flexure can include any component with sufficient tensile strength to bear the load placed on it from use of the implant **100** and sufficient flexibility to provide the desired range of motion along each axis. In some embodiments, the flexure is configured to eliminate stress concentrations and plastic deformation along its length. In some cases, this ensures that the flexure bends at stress levels that are well below its yield strength. Selecting geometries that are tailored to the material, the desired performance stiffness, and the desired range of motion is, in accordance with some embodiments, enabled by this flexure. For example, the flexure can include a thin sheet of material (e.g., metal, plastic, carbon fiber, polymer material, fabric, ceramic, natural material, synthetic material, or any other material), or it can include one or more strands, cords, sheets, walls, rods, hooks, chains, bars, filaments, or any other tension-bearing components that can bend well below their yield stress. By way of non-limiting illustration, FIGS. 7, 11, and 12 show flexures **114** that include relatively straight and relatively narrow sheets of material supported by supports **116a** and **116b** on either end.

[0070] In some embodiments, the flexibility of the flexure assembly **110** can be adjusted by adjusting the flexure **114**, including by adjusting the material the flexure comprises, by adjusting the shape of the flexure, by adjusting the dimensions of the flexure, by perforating the flexure, or in any other suitable manner. For example, a flexure **114** with a narrow length and width (e.g., a large height to length or height to width aspect ratio, as shown in FIG. 5) can be flexible along each axis, but that flexibility can be decreased along one or more axes by increasing the length or width of the flexure **114** (as seen in FIGS. 6A-6D in comparison to FIG. 5). In some embodiments, the flexibility of the flexure is tailored to achieve the desired flexibility of the flexure assembly (as discussed above) to consequently achieve the desired flexibility of the implant **100** as a whole.

[0071] Although the flexure **114** can be positioned anywhere within the flexure assembly **110**, some embodiments include at least a portion of the flexure in the center of mass of the flexure assembly for added stability. In some cases, the flexure is positioned such that all portions of the flexure receive an approximately equal share of the total load. In this regard, in accordance with some embodiments, placing the center of rotation of the device at the spinal level's center of rotation helps provide for anatomical motion. Furthermore, in some embodiments, the flexure assembly provides for a moving center of rotation that approximates the helical axis of motion of the spine level.

[0072] The flexure **114** can have any dimensions that allow it to have sufficient tensile strength to support the load placed on the implant **100** and to have the desired flexibility along each axis.

Generally speaking, the aspect ratio of some embodiments of the flexure can be much different than the aspect ratio required for a compression-based stem. For example, the flexure can have a much greater height-to-width or height-to-length ratio (or both) without danger of buckling or breaking.

[0073] According to some embodiments, the flexure **114** has a height between 1% and 100% of the height of the flexure assembly **110** (or any subrange thereof, such as 50%±10%, 60%±10%, 70%±10%, 80%±10%, or 90%±10%). That said, some embodiments of the flexure are shorter than the flexure assembly (e.g., the supports extend farther than the flexure on either side of the flexure's height) to ensure that the flexure is placed in tension (and not compression) when the implant **100** is loaded. In some embodiments, at least one of the width and the length of the flexure is substantially less than the height of the flexure, such as less than 50% of the height (or any other number less than 50%, such as less than 40%, 30%, 25%, 20%, 15%, 10%, 5%, or even 1% of the height). That said, the length of the flexure can be any length up to 100% of the length of the flexure assembly, and the width can be any width up to 100% of the width of the flexure assembly (although this would require an alternative configuration of the supports in some cases, which in some embodiments are situated on either side of the flexure, thereby adding to the width of the flexure assembly). In short, the flexure can be customized by modifying the height-to-length-to-width aspect ratio as needed to form a flexure with the desired flexibility along the proper axes. Thus, the aspect ratio can be any ratio of h-l-w (where h is the height, l is the length, and w is the width) with h being any number between 1 and 100 (or any subrange thereof), l being any number between 1 and 100 (or any subrange thereof), and w being any number between 1 and 100 (or any subrange thereof), but with h being at least two times at least one of/and w. As examples: 10-1-1, 10-5-2, 10-10-1, 15-2-1, 8-9-1, and 12-1-8 are all possible h-l-w aspect ratios that may be useful for a flexure. In some embodiments, h is at least three times at least one of l and w, at least four times at least one of l and w, at least five times at least one of l and w, or otherwise substantially larger than at least one of l and w. In some embodiments, the aspect ratio of the flexure is configured to ensure that the flexure assembly has a desired flexibility along one or more axes, as described above.

[0074] The flexure **114** can also be any shape suitable for being placed under tension. For example, some embodiments of the flexure include one or more substantially linear, planar, cylindrical, columnar (e.g., cylindrical-columnar, square-columnar, triangular-columnar, polygonal-columnar, or otherwise configured in a column), tubular, sheet-like, undulated, coiled, wavy, or any other suitable shaped components. By way of non-limiting illustration: FIG. 5 shows an embodiment of a substantially linear rectangular-columnar flexure **114**; FIG. 6A shows an embodiment of a substantially planar rectangular-columnar flexure **114**; and FIG. 7 shows an embodiment of a substantially planar rectangular-columnar flexure.

[0075] It should be noted that a normal human disc typically resists compression, is relatively flexible in flexion/extension, is less flexible in lateral bending, and is even less flexible in axial rotation. Also, a human disc largely resists relative translation between the vertebral endplates. In this regard, getting each relative stiffness right at the same time, without sliding contact, and with a proper center of rotation is a feature of some embodiments of the described implant **100-a** feature that is not found in most, if any, competing devices. Moreover, providing all that with a graft window for nutrient flow is also a feature of some embodiments of the described implant-again something that is not found in most, if any, competing devices.

[0076] Where the flexure assembly **110** includes one or more supports **116**, the supports can include any component allowing load to be transferred from the loading regions **112** to the flexure **114** in such a manner as to place the flexure under tension. Some embodiments include at least two supports, such that the first support **116a** can transfer the load from a first loading region **112a** to a first end **114a** of the flexure disposed opposite the first loading region, and the second support **116b** can transfer the load from a second loading region **112b** to a second end of the flexure **114b**.

disposed opposite the second loading region (as shown in FIG. 4). This contrasts with a compression-based stem (as shown in FIG. 3) in which the load from a first loading region **62a** is transferred directly to a first end **64a** of the stem and the load from a second loading region **62b** is transferred directly to a second end **64b** of the stem, such that the stem is placed under compression.

[0077] The supports **116** can include any suitable components that provide the desired support, including one or more pillars, columns, rods, walls, sheets, beams, posts, bars, or other supports. The supports can have any supportive shape, such as cylindrical, conical, square-beam-shaped, V-shaped, triangular, sheetlike, or otherwise shaped. By way of non-limiting illustration, FIG. 4 shows an embodiment comprising pillar-like supports **116a** and **116b**. FIGS. 5-6D show embodiments comprising sheet-like supports **116a** and **116b**. FIG. 7 shows a first flexure assembly **110a** with contoured supports (e.g., thicker near loading regions **112a** and **112b**, and thinner near the flexure **114**, with curved contours joining the supports to the flexure). FIG. 12 shows an embodiment having a first support **116a** and a second support **116b**, each having a sloped surface to provide additional support near the loading regions **112a** and **112b**. FIG. 13 shows an embodiment comprising multiple flexure assemblies **110a**, **110b**, **110c**, and **110d** with supports **116a** and **116b** that have a general rectangular block configuration. FIG. 14 shows an embodiment comprising a second flexure assembly **110b** with slanted supports **116a** and **116b**.

[0078] The supports **116** can also have any suitable dimensions allowing them to support a tension-based flexure **114**. For example, the supports can have any suitable length, but some embodiments of the supports have a length approximately equal to the length of the flexure assembly (e.g., 90% $\pm$ 10% of the flexure assembly length). Some embodiments have a length of approximately between 1% and 15% of a length of the flexure assembly. The supports can also have any suitable width, but in some embodiments the width is sufficient to ensure that the supports are substantially rigid (e.g., having a flexibility of less than 10% of the overall flexibility of the flexure assembly **110**, a flexibility of less than 1 degree, or otherwise having a rigidity sufficient to prevent buckling and breaking of the flexure assembly when placed under a load as required by the implant **100**). Some embodiments of the supports have a width between 1 mm and 10 mm, or any subrange thereof. Some embodiments of the supports have a width between 1% and 10% of the width of the flexure assembly, or any subrange thereof. The supports can also have any suitable height that allows them to transfer a compressive force received at one end (the end by the loading region) to a tensile force at the other end (through its coupling with the flexure **114**). For example, in some embodiments the supports have a height that is greater than a height of the flexure. In some cases, the supports have a height between 1% and 20% greater than the height of the flexure (or any subrange thereof, such as 5% $\pm$ 3%), and in some cases, the supports have a height between 0.1 mm and 20 mm greater than the height of the flexure. In some embodiments, the supports have a height of between 50% and 100% of the height of the flexure assembly (or any subrange thereof, such as 85% $\pm$ 10%).

[0079] In some embodiments, the supports **116** are offset from one another in one or more directions. For example, some embodiments of the supports are offset (e.g., by up to 50% or any percentage below that, such as 25%, 10%, 5%, 1%, or another percentage) along the axis of the length of the flexure assembly **110** or the axis of the height of the flexure assembly (or both). By way of non-limiting illustration, FIG. 5 shows an embodiment in which the supports **116a** and **116b** are offset along the length **1** and height **h** of the flexure assembly **110**. In many cases, the supports are naturally spaced apart along the width **w** (e.g., offset by between 100% and 1000%, or any subrange thereof), as this allows one support to be positioned on either side of the flexure **114**.

[0080] The flexure **114** and supports **116** can couple together in any suitable manner. For example, in some embodiments, the flexure and support are joined by one or more intermediate components **115**. In some embodiments, at least one of the flexure and the supports is contoured such that they join together directly. Whether the supports include an intermediate component or portion or are contoured to join to the flexure directly, they can include any suitable feature, such as one or more

arches, squared couplers, horizontal couplers, diagonal couplers, semi-circles, bends, curves, processes, protrusions, overhangs, ledges, outcroppings, or any other suitable contours or components allowing them to join to the flexure. By way of non-limiting illustration, FIG. 5 shows an embodiment of the flexure **114** joined to supports **116a** and **116b** via one or more intermediate components **115a** and **115b**. In contrast, FIG. 7 shows flexure assemblies **110a** and **110b** having contoured supports **116a** and **116b** that couple directly to flexures **114**.

[0081] Although the loading regions **112**, flexure **114**, and supports **116** can be separate components, some embodiments of the flexure assembly **110** are integrally formed as a single piece. For example, in some cases, at least the flexure and the supports are integrally formed as a single piece. Where one or more components of the flexure assembly are provided as a separate piece, the components can be joined together in any suitable manner, such as via one or more nails, screws, bolts, staples, eyelets, magnets, adhesives, welds, interference fits, friction fits, mechanical engagements, tongue-and-groove connections, snaps, ties, rivets, pins, fasteners, stakes, wire ties, or any other coupling mechanisms.

[0082] Although some embodiments of the implant **100** include only a single flexure assembly **110**, some embodiments include multiple flexure assemblies, such as a first flexure assembly **110a**, a second flexure assembly **110b**, a third flexure assembly **110c**, a fourth flexure assembly **110d**, or any other number of additional flexure assemblies. In this regard, some embodiments of the flexure assembly are more flexible (or less flexible) along one or more particular axes than are other embodiments. Accordingly, coupling such embodiments together can result in a device with the desired flexibility and the desired strength along any or all axes. Multiple flexure assemblies can also operate independently of one another, thereby adding flexibility and stability capabilities to multiple portions of the implant.

[0083] Where multiple flexure assemblies **110** are used together, they can be coupled or otherwise arranged in any suitable manner, such as by being stacked (e.g., arranged in series), placed end to end, placed side-by-side (e.g., arranged in parallel), being offset from each other, or placed in any other suitable configuration. By way of non-limiting illustration, FIGS. 6A and 6B show some embodiments depicting flexure assemblies **110** in a series arrangement, whereas FIGS. 6C and 6D show some embodiments having a parallel arrangement. In some embodiments, multiple flexure assemblies may be arranged side-by-side, but coupled together in such a manner so as to cause the load path to transmit at least a portion of the load (and in some cases, the whole load) from one flexure to another flexure in series. Thus, multiple flexure assemblies can operate entirely in series, entirely in parallel, or partially in series and partially in parallel. By way of non-limiting illustration, FIG. 12 shows an embodiment of the implant **100** with a first flexure assembly **110a** coupled to a second flexure assembly **110b** such that a portion of the load path is transferred from one flexure assembly to the other in series, while the flexure assemblies still each take some of the load on their own, thereby operating partly in parallel.

[0084] Where multiple flexure assemblies **110** are used, the flexure assemblies can be placed at any angle or orientation with respect to each other. For example, some embodiments of the flexure assemblies have the same orientation along every axis, but some embodiments have at least one flexure assembly that is rotated with respect to at least one other flexure assembly along at least one axis in any amount between 0 degrees and 360 degrees, or any subrange thereof. As an example, one flexure assembly is rotated between 15 degrees and 95 degrees, between 35 degrees and 55 degrees, or any other suitable range of rotation, along the torsion axis with respect to another flexure assembly. As another example, one flexure assembly can be rotated 45 degrees (± 10 degrees) along the flexion/extension axis as well as 120 degrees (± 10 degrees) along the lateral bending axis. As another example, one flexure assembly can be rotated 180 degrees (± 10 degrees) of another flexure assembly along one or more axes, and a third flexure assembly can be rotated in any amount along any axis with respect to either or both of the other two flexure assemblies. By way of non-limiting illustration, FIG. 6A shows an embodiment of the first flexure

assembly **110a** rotated between 10 degrees and 40 degrees along a torsion axis with respect to a second flexure assembly **110b**. FIGS. **6B-6D** show a first flexure assembly **110a** rotated approximately 45 degrees (± 5 degrees) along a torsion axis with respect to a second flexure assembly **110b**.

[0085] Furthermore, in some embodiments, multiple flexure assemblies **110** are aligned with each other, and in some embodiments, multiple flexure assemblies are offset from one another (in any suitable amount, such as anywhere between 0% and 500%, or any subrange thereof). For example, in some embodiments, a first flexure assembly is positioned next to and aligned with a second flexure assembly such that a face of the first flexure assembly is flush with a face of the second flexure assembly. As another example, in some embodiments with multiple flexure assemblies, the flexure assemblies are slightly offset (e.g., between 1% and 5% offset, between 2% and 10% offset, between 5% and 20% offset, or otherwise offset) from each other such that one flexure assembly is higher, farther forward, farther backward, or to farther to a side than the other one. In some embodiments, the flexure assemblies are greatly offset such that they do not align at all.

[0086] In some embodiments, the rotation, position, or other configuration of one or more flexure assemblies **110** is selectively adjustable or otherwise customizable. In this regard, the flexure assembly can be adjustable or customizable in any suitable manner, including by being specifically designed for a patient based on the patient's particular needs (e.g., as determined by X-ray, imaging, or in any other suitable manner), having the endplates **102** be configured to couple with the flexure assemblies in one of multiple orientations, through the use of one more adjustment mechanisms, or in any other suitable manner.

[0087] In some embodiments, multiple flexure assemblies **110** are coupled together. Where multiple flexure assemblies are coupled together, they can be coupled in any suitable manner, such as through welding, being formed as an integral part, adhering together using one or more adhesives, via one or more couplers **118**, via one or more fasteners, via one or more frictional or mechanical engagements, or through any other suitable manner. Where a coupler is used, the coupler can include any suitable selective, permanent, or semi-permanent coupling component, such as one or more one or more nails, pins, screws, bolts, rods, sheets, staples, eyelets, magnets, hook-and-loop fasteners, adhesives, welds, interference fits, friction fits, tongue-and-groove connections, snaps, ties, rivets, wire ties, or any other type of coupler. In some cases, the coupler itself includes one or more flexure assemblies or parts thereof (e.g., having any of the properties relating to flexure assemblies as discussed herein), thereby allowing the coupler to have enhanced stability and flexibility. The coupler can also have any suitable dimensions (e.g., any width, height, and length suitable for coupling multiple flexure assemblies together). By way of non-limiting illustration, FIG. **6D** shows an embodiment of a first flexure assembly **110a** and a second flexure assembly **110b** coupled together via a coupler **118** that includes a flexible piece of material (or any other suitable material) coupled to both flexure assemblies. FIGS. **8-10** show some embodiments of implants **100** with multiple flexure assemblies **110a** and **110b** coupled together via one or more couplers **118**, where such coupler or couplers includes a thin, flexible piece of material (or any other suitable material) joining a first support **116a** of a first flexure assembly **110a** to a second support **116b** of a second flexure assembly **110b** such that the respective flexure assemblies are perpendicular with respect to each other. FIG. **12** shows a pair of flexure assemblies **110a** and **110b** that are integrally joined together, such that a coupler **118** is substantially in the form of an extension of the flexure assemblies allowing them to merge together into a unified construct. FIG. **13** shows an X-shaped coupler **118** configured to join four flexure assemblies **110a**, **110b**, **110c**, and **110d** together.

[0088] In some embodiments with multiple flexure assemblies **110**, a first flexure assembly **110a** can primarily govern flexibility along the lateral bending axis, and a second flexure assembly **110b** can primarily govern flexibility along the flexion/extension axis (as shown in FIGS. **11-12**), or vice versa. In some embodiments with multiple flexure assemblies, all such flexure assemblies are

configured to sum together to provide the desired flexibility along any particular axis (e.g., the flexion/extension axis, the lateral bending axis, or the torsion axis).

[0089] Where multiple flexure assemblies **110** are used, the flexure assemblies can have any size with respect to each other. For example, in some embodiments, a first flexure assembly **110a** has a height between 50% and 200% of a height of a second flexure assembly **110b**, or any subrange thereof. For example, in some cases, the first flexure assembly's height is 95%±4% the height of the second flexure assembly (or 92%±5%, 105%±5%, 110%±5%, 65%±15%, or any other suitable height differential). By way of non-limiting illustration, FIG. **12** shows an embodiment of a first flexure assembly **110a** that is slightly shorter than a second flexure assembly **110b**, such that a first endplate **102a** and a second endplate **102b** are closer together near the first flexure assembly than near the second flexure assembly. Similarly, either or both of the width or the length of one flexure assembly can be anywhere between 20% and 1000% (or any subrange thereof) of the width or length of another flexure assembly. For example, in some embodiments, a first flexure assembly **110a** has a length (or width) that is 200%±50% (or 110%±5%, 150%±15%, 250%±50%, or any other suitable percentage range) the length (or width) of a second flexure assembly. By way of non-limiting illustration, FIGS. **7** and **8** show embodiments of a first flexure assembly **110a** that has a length that is greater than a length of a second flexure assembly **110b**, wherein the first and second flexure assemblies have a substantially similar width.

[0090] Some embodiments include one or more additional elements to modulate any of the properties of the described implant **100**. For example, some embodiments include one or more additional elements for increasing the relative stiffness of the implant, such as one or more reinforcement elements **122**. Reinforcement elements can include one or more springs, coils, coil packs, beams, bars rods, struts, posts, torsional elements, frame, lattices, resilient members, scaffolds, porous structures, trusses, supports, fills, woven supports, twisted supports, interconnecting supports, matrices, compressive elements, or any other additional elements. In some embodiments, an array **120** of reinforcement elements is provided.

[0091] In some embodiments, the array **120** of reinforcement elements **122** forms a porous structure. The porous structure can include any structure defining gaps, spaces, air pockets, voids, cavities, hollows, or other pores capable of promoting bone growth (e.g., cortical bone growth, trabecular bone growth, or other bone growth).

[0092] The reinforcement elements **122** can include any component capable of increasing the relative stiffness, strength, compliance, or integrity of the implant **100**, of exerting a compressive load, of encouraging bone growth, or of modulating any of the other properties of the implant (e.g., flexibility along any particular axis). For example, the reinforcement elements can include one or more supports, meshes, grids, strands, filaments, foams, lattices, matrices, sponge-shaped supports, supports that resemble cancellous or trabecular bone, wires, grates, springs, coils, frames, frets, screens, reticulations, or other auxiliary components. By way of non-limiting illustration, FIGS. **14-15** show some embodiments of the implant **100** having an array **120** formed of a plurality of compression-based reinforcement elements **122**.

[0093] In some embodiments, the reinforcement elements **122** include at least a first reinforcement element **122a** and a second reinforcement element **122b** (and any number of additional reinforcement elements). Where the implant **100** includes multiple reinforcement elements **122** (e.g., an array **120** of such elements), the reinforcement elements can be arranged in any manner suitable for achieving their desired functions. For example, in some embodiments, reinforcement elements are disposed in one or more portions of the implant not occupied by flexure assemblies **110**. In some embodiments, reinforcement elements are disposed around all or part of a perimeter of the implant (such as along one or more of a dorsal face, a ventral face, a left face, or a right face). In some embodiments, one or more reinforcement elements are disposed within an interior of the implant (e.g., adjacent to, surrounding, filling, flanking, or between one or more flexure assemblies). In some embodiments with multiple reinforcement elements, the reinforcement

elements are interlaced, interwoven, interlocking, coupled together, included in a repeating pattern, or included in a mirrored pattern (which in some cases makes the reinforcement elements easier to manufacture or adds additional strength), but the reinforcement elements may also be included in any other pattern or arrangement. By way of non-limiting illustration, FIG. **14** shows an embodiment of the implant **100** with an array **120** of reinforcement elements **122**, wherein the reinforcement elements **122** include coils, and wherein a first reinforcement element **122a** includes a right-hand (clockwise) coil and a second reinforcement element **122b** includes a left-hand (counterclockwise) coil, thus lending the array a repeating mirrored pattern around a portion of a perimeter of the implant. In some embodiments, one or more coils are separate and discrete from one or more adjacent coils. In some embodiments, however, one or more coils are coupled to, integrally formed with, tangentially coupled to, or otherwise connected with one or more adjacent coils.

[0094] In some embodiments, the reinforcement elements **122** can couple to any suitable portion of the implant. Indeed, in some embodiments, the reinforcement elements are coupled to the endplates **102** (or to any other suitable component, such as one or more flanges **104** or flexure assemblies **110**) of the implant **100**. While this coupling can be done in any suitable manner, in some embodiments, it is done via one or more couplers (e.g., one or more nails, screws, bolts, staples, eyelets, magnets, adhesives, welds, interference fits, friction fits, tongue-and-groove connections, snaps, ties, rivets, stakes, wire ties, catches, frictional engagements, mechanical engagements, or any other suitable couplers), and in some embodiments the reinforcement elements are integrally formed with the endplates or other components of the implant (e.g., via additive manufacturing, 3D-printing, molding, etching, subtractive manufacturing, or in any other suitable manner). In some cases, a flexure assembly **110** that can be printed with additive manufacturing includes one or more novel shapes, such as support structures while the material is cooling so that the geometry does not sag, warp, or otherwise deform. In some embodiments, electropolishing is used, for instance, to improve fatigue performance or for any other suitable purposes. In some cases, however, the chemicals used in electropolishing can be cytotoxic. Accordingly, some embodiments of the described systems and methods include a flexure geometry that allows for post-polish cleaning (e.g., to allow all such chemicals to be removed). Indeed, in some cases, blind cavities, small voids, and other difficult to clean geometries are avoided.

[0095] Some embodiments of the implant **100** define one or more cavities **124**. The cavity can include any suitable aperture, passage, recess, indentation, space, orifice, or other cavity configured to receive a material, such as a biological packing material. Where a biological material for packing into the implant is used, it can include any suitable biological material, such as one or more of ground bone, calcium phosphate, bone graft filler, or other packing material used in connection with spinal surgeries. In some cases, the material is granular, and in some cases the granules are sized so as to resist dissociation from the implant in large quantities (e.g., the granules are larger than the “pore size” of the gaps between coils of reinforcement members **122**).

[0096] In some embodiments, one or more of the endplates **102**, flexure assemblies **110**, or other components of the implant **100** optionally include (or define) one or more windows **126** through which a cavity **124** can be accessed (e.g., through which biological material can be inserted into the cavity). Where a window is included, the window can be any suitable size (e.g., between 10% and 80% of the size of surface area of an endplate **102**, or any subrange thereof) and any suitable shape (e.g., circular, semi-circular, square, rectangular, triangular, pentagonal, hexagonal, trapezoidal, polygonal, symmetrical, asymmetrical, or otherwise regularly or irregularly shaped). In some embodiments, all or part of the window is defined by one or more components of the implant, such as one or more portions of an endplate, one or more loading regions **112** of a flexure assembly **110**, one or more reinforcement components **122**, or any other components of the implant. By way of non-limiting illustration, FIG. **9** shows an embodiment of a window **124** formed via an L-shaped portion of a first endplate **102a** and a first loading region **112a** of a second flexure assembly **110b**,

the window being substantially rectangular (with rounded edges or corners) and having a size of less than 25%±15% of the external surface area of the first endplate, the window providing access to a cavity **124** disposed between the first endplate and a second endplate **102b**, and the cavity being configured to receive a biological packing material.

[0097] According to some embodiments of the disclosed systems and methods, a method of providing the spinal implant **100** is provided. In some embodiments, the method includes forming the spinal implant. The spinal implant can be formed in any suitable manner, such as through casting (e.g., continuous casting, die casting, mold casting, resin casting, sand casting, or any other suitable type of casting), molding (e.g., metallurgy, compaction, injection molding, or any other suitable type of molding), forming (e.g., forging, extrusion, pressing, or any other suitable type of forming), machining (e.g., milling, turning, drilling, or any other suitable type of machining), joining (e.g., welding, ultrasonic welding, or any other suitable type of joining), additive manufacturing (e.g., 3D printing, deposition, layered manufacturing, or other additive manufacturing), subtractive manufacturing, suitable type of, or in any other suitable manner. For example, some embodiments of the implant are formed via additive manufacturing (e.g., 3D printing) in a manner that allows multiple (or all) components of the implant to be formed as a single integral part. That said, in some embodiments, one or more components are formed separately and coupled to one or more other components of the implant. In such cases, such coupling can be done in any suitable manner, such as via one or more anchors, adhesives, welds, magnets, interference fits, friction fits, mechanical engagements, tongue-and-groove connections, snaps, ties, rivets, stakes, wire ties, pins, catches, clamps, threaded engagements, or any other suitable couplers.

[0098] According to some embodiments, the method includes forming one or more components of the implant **100** to specific specifications (e.g., as discussed above). For example, in some embodiments, one or more components are formed to meet a certain threshold of flexibility, stability, stiffness, or any other suitable attribute.

[0099] In some embodiments, the method includes identifying desired parameters for the implant **100** to satisfy. Such parameters can include any suitable parameters, such as flexibility (either in general or along one or more specific axes), stiffness, size, shape, or other aspects of the implant. In some cases, the desired parameters are patient-specific, spinal-segment-specific, or otherwise dependent on the desired use case. Moreover, such parameters can be identified based on any suitable information, such as patient demographics, imaging, patient symptoms (or other patient-provided information), test results, bone density information, bone shape information, or any other suitable information. By way of non-limiting illustration, some embodiments of the method include performing pre-operative imaging to identify desired parameters.

[0100] Where the method includes pre-operative imaging, the imaging can include any suitable type of imaging, such as X-ray, CT Scan, MRI (or any variant thereof, such as fMRI), ultrasound, or any other imaging technique. By way of non-limiting illustration, in some embodiments, the parameters are determined based on water content as measured via MRI, native flexibility as measured via functional X-rays, or other information obtained through imaging. In this regard, unlike many (if not all competing systems and methods), some embodiments of the described systems and methods include matching the index level to the adjacent levels in terms of radiographic height, stiffness, and range of motion. In this regard, using a simple series of flexion/extension X-rays add little to no additional burden in pre-operative imaging.

[0101] Some embodiments of the implant **100** are configured to be custom tailored to a patient's needs. Accordingly, some embodiments of the method include tailoring the implant to a patient's needs. Such tailoring can be done in any suitable manner, such as prior to (or during) formation of the implant or any of its constituent components, or by making adjustments after the formation of the implant. In some embodiments where adjustments are made after the formation of the device, such adjustments can include any suitable adjustments to the dimensions, shape, orientation, or

other configuration of any of the components. For example, suitable adjustments can include: adding or removing material (e.g., to make one or more flexure **114** more or less stiff, removing one or more reinforcement elements **122**, etc.); adjusting the position of one or more components (e.g., rotating a flexure assembly **110** to adjust the flexibility along certain axes); reshaping one or more components (e.g., to provide a more even load path through one or more flexure assemblies); increasing or decreasing the stiffness of a support **116**, a flexure, a coupler **118**, or another component (e.g., by changing the length, width, or height of such component, or by changing its material composition, or by changing its general shape or configuration); limiting flexibility through a stop element (e.g., a contact-enabled stop); or otherwise adjusting one or more components or parameters of the implant. In any case, it should be noted that the size and shape of primary fixation does not necessarily preclude multiple level implantations.

[0102] The systems and methods discussed herein can be modified in any suitable manner. To illustrate, some embodiments include multiple flexure assemblies **110** having a specific configuration. For example, the flexure assemblies of some embodiments are generally arranged in a X shape, a T shape, a plus shape, a cross shape, an H shape, a V shape, or another configuration. In this regard, an X shape or another shape with centrally concentrated components may be particularly useful in some cases, because resistance to torsion may be reduced through minimization of outer fibers. In this regard, torsion resistance can, in some embodiments, be stronger when there is more material disposed farther away from an axis of rotation (e.g., a rod with a large diameter is more difficult to twist than a rod with a small diameter, even if the amount of material used is the same). By way of non-limiting illustration, FIG. **13** shows an embodiment of the implant **100** with an X-shaped arrangement of flexure assemblies **110a**, **110b**, **110c**, and **110d**, thereby reducing torsional stiffness.

[0103] Any and all of the components in the figures, embodiments, implementations, instances, cases, methods, applications, iterations, examples, elements, and other parts of this disclosure and the implant **100** can be combined in any suitable manner. Additionally, any component can be removed, separated from other components, modified with or without modification of like components, or otherwise altered together or separately from anything else disclosed herein.

[0104] As used herein, the singular forms “a”, “an”, “the” and other singular references include plural referents, and plural references include the singular, unless the context clearly dictates otherwise. For example, reference to a flexure (or “the” flexure) includes reference to one or more flexures, and reference to supports includes reference to one or more supports. In addition, where reference is made to a list of elements (e.g., elements a, b, and c), such reference is intended to include any one of the listed elements by itself, any combination of less than all of the listed elements, and/or a combination of all of the listed elements. Moreover, the term “or” by itself is not exclusive (and therefore may be interpreted to mean “and/or”) unless the context clearly dictates otherwise. Similarly, the term “and” by itself is not exclusive (and therefore may be interpreted to mean “and/or”) unless the context clearly dictates otherwise. Furthermore, the terms “including”, “having”, “such as”, “for example”, “e.g.”, and any similar terms are not intended to limit the disclosure, and may be interpreted as being followed by the words “without limitation”.

[0105] In addition, as the terms “on”, “disposed on”, “attached to”, “connected to”, “coupled to”, etc. are used herein, one object (e.g., a material, element, structure, member, etc.) can be on, disposed on, attached to, connected to, or otherwise coupled to another object—regardless of whether the one object is directly on, attached, connected, or coupled to the other object, or whether there are one or more intervening objects between the one object and the other object. Also, directions (e.g., “front”, “back”, “on top of”, “below”, “above”, “top”, “bottom”, “side”, “up”, “down”, “under”, “over”, “upper”, “lower”, “lateral”, “right-side”, “left-side”, “base”, etc.), if provided, are relative and provided solely by way of example and for ease of illustration and discussion and not by way of limitation.

[0106] The described systems and methods may be embodied in other specific forms without

departing from their spirit or essential characteristics. The described embodiments, examples, and illustrations are to be considered in all respects only as illustrative and not restrictive. The scope of the described systems and methods is, therefore, indicated by the appended claims rather than by the foregoing description. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope. Moreover, any component and characteristic from any embodiments, examples, and illustrations set forth herein can be combined in any suitable manner with any other components or characteristics from one or more other embodiments, examples, and illustrations described herein.

Claims

1. An implant for standalone interbody fusion, the implant comprising: a first endplate; a second endplate; and a load-bearing component disposed between the first endplate and the second endplate, wherein the load-bearing component has a height that is at least twice as great as at least one of a width and a length of the load-bearing component.
2. The implant of claim 1, wherein at least one of the first endplate and the second endplate comprises a flange, the flange comprising a primary fixation passage configured to receive an anchor for coupling at least one of the first endplate and the second endplate to a spinal segment of a patient.
3. The implant of claim 2, wherein the flange further comprises a secondary fixation passage configured to receive a locking cap, wherein the locking cap is configured to help prevent the anchor from backing out of the primary fixation passage.
4. The implant of claim 1, wherein the load-bearing component comprises a first flexure, and wherein a portion of the first flexure is configured to be placed under tension when a compressive force is applied to at least one of the first endplate and the second endplate.
5. The implant of claim 4, further comprising a flexure assembly comprising: a first loading region configured to receive a first portion of the compressive force; a second loading region configured to receive a second portion of the compressive force; the first flexure; a first support configured to transfer the first portion of the compressive force to a first end of the first flexure, wherein the first end of the first flexure is positioned opposite the first loading region; and a second support configured to transfer the second portion of the compressive force to a second end of the first flexure, wherein the second end of the first flexure is positioned opposite the second loading region, and wherein a transfer of the compressive force from the first loading region and the second loading region causes the first flexure to support the compressive force with an opposing tensile force.
6. The implant of claim 5, further comprising a second flexure assembly, wherein the second flexure assembly comprises a second flexure.
7. The implant of claim 6, wherein the first flexure is configured to be flexible along a flexion/extension axis, and wherein the second flexure is configured to be flexible along a lateral bending axis.
8. The implant of claim 1, wherein the implant is flexible along each of a flexion/extension axis, a lateral bending axis, and a torsion axis.
9. The implant of claim 8, wherein the implant is configured to stabilize motion to a physiological range along each of the flexion/extension axis, the lateral bending axis, and the torsion axis.
10. The implant of claim 1, wherein the load-bearing component is configured to withstand at least 25% of a compression force applied to the first endplate and the second endplate without buckling and without breaking when the compression force results from compression of the implant between a first vertebra and a second vertebra of a patient's spine.
11. The implant of claim 1, further comprising a reinforcement element configured to further stabilize the implant.

- 12.** The implant of claim 11, wherein the reinforcement element is included in an array of reinforcement elements forming a porous structure.
- 13.** A flexure-based standalone interbody fusion implant comprising: a first flexure assembly comprising: a first loading region configured to receive a first portion of a compressive force; a second loading region configured to receive a second portion of the compressive force; a first support associated with the first loading region; a second support associated with the second loading region; and a flexure coupled to the first support and the second support, wherein a portion of the flexure is configured to be placed under tension in response to the compressive force being applied to at least one of the first loading region and the second loading region.
- 14.** The flexure-based standalone interbody fusion implant of claim 13, wherein at least one of a stiffness of the implant along a flexion/extension axis, a stiffness of the implant along a lateral bending axis, and a stiffness of the implant along a torsion axis is selectively adjustable.
- 15.** The flexure-based standalone interbody fusion implant of claim 13, wherein a flexibility of the implant along a flexion/extension axis is $7\text{ degrees}\pm 4\text{ degrees}$, a flexibility of the implant along a lateral bending axis is $5\text{ degrees}\pm 3\text{ degrees}$, and a flexibility of the implant along a torsion axis is $4\text{ degrees}\pm 4\text{ degrees}$.
- 16.** The flexure-based standalone interbody fusion implant of claim 13, further comprising a second flexure assembly.
- 17.** The flexure-based standalone interbody fusion implant of claim 16, wherein the second flexure assembly is coupled to the first flexure assembly via a coupler.
- 18.** The flexure-based standalone interbody fusion implant of claim 16, wherein the first flexure assembly has a first orientation and the second flexure assembly has a second orientation, and wherein the first orientation is different from the second orientation.
- 19.** A method of providing an implant, the method comprising: obtaining a first flexure assembly comprising: a first loading region configured to receive a first portion of a compressive force; a second loading region configured to receive a second portion of the compressive force; a first support associated with the first loading region; a second support associated with the second loading region; and a flexure coupled to the first support and the second support, wherein the flexure is configured to be placed under tension in response to the compressive force being applied to at least one of the first loading region and the second loading region; and inserting the implant into an interbody space of a spine of a patient such that adjacent vertebrae of the spine apply the compressive force to the implant.
- 20.** The method of claim 19, further comprising configuring at least one of the first support, the second support, and the flexure to provide a patient-specific flexibility along at least one of a flexion/extension axis, a lateral bending axis, and a torsion axis.
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